

RAYONG, Thailand

WMO: 484780

Lat: 12.6339N

Lon: 101.3400E

Elev: 5

StdP: 101.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	20.8	12.1	8.8	23.3	14.0	10.0	24.4	5.5	29.2	5.2	29.4	1.3	350	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	5.9	35.1	27.9	34.2	27.5	33.6	27.2	28.8	33.5	28.5	32.8	28.1	32.2	1.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.7	23.7	31.7	27.2	23.0	31.2	27.0	22.8	31.0	93.9	33.5	92.0	32.9	90.4	32.4	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.3	4.8	17.4	36.0	1.6	1.9	16.2	37.4	15.3	38.5	14.4	39.5	13.3	40.9		
(5)			14.1	29.5	1.8	1.0	12.8	30.2	11.8	30.8	10.8	31.3	9.5	32.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	26.7	28.1	29.3	30.3	30.1	29.7	29.3	29.1	28.5	28.3	28.0	26.9		
	DBStd	1.76	1.58	1.59	1.56	1.56	1.39	1.22	1.21	1.08	1.14	1.01	1.16	1.73		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6821	518	507	599	610	622	590	597	592	556	566	540	524		
	CDD18.3	3779	259	274	341	360	364	340	339	333	306	308	290	266		
	CDH23.3	45213	2599	3086	4225	4810	4884	4469	4301	4167	3602	3322	3087	2660		
	CDH26.7	19556	939	1194	1900	2490	2458	2121	1887	1758	1371	1193	1212	1032		
(14) Wind	WSAvg	1.8	1.4	1.7	1.8	1.8	1.9	2.4	2.4	2.4	1.7	1.2	1.4	1.5		
(15) Precipitation	PrecAvg	1475	29	31	69	80	191	188	186	146	280	212	48	8		
	PrecMax	1974	112	112	284	226	398	337	344	342	481	404	111	39		
	PrecMin	987	0	1	2	65	12	46	47	115	60	0	0	0		
	PrecStd	239	33	26	61	53	88	85	100	65	84	74	36	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	34.6	35.6	36.7	35.7	35.1	35.0	34.2	34.0	34.2	34.2	34.6		
		MCWB	26.3	27.4	27.7	28.9	28.5	27.8	28.0	27.4	27.5	27.2	26.5	26.4		
	2%	DB	32.4	33.1	34.1	35.0	34.2	33.7	33.2	33.0	32.7	33.0	33.2	33.1		
		MCWB	25.0	26.1	27.4	28.1	28.0	27.4	27.2	26.8	26.9	26.6	25.8	25.1		
	5%	DB	31.6	32.0	33.0	33.9	33.3	32.7	32.1	32.0	31.8	32.1	32.5	32.0		
		MCWB	24.6	25.7	27.0	27.8	27.7	27.2	26.8	26.7	26.6	26.4	25.4	24.6		
	10%	DB	30.7	31.1	31.9	33.0	32.6	31.9	31.2	31.0	30.9	31.2	31.7	31.1		
		MCWB	24.5	25.6	26.8	27.6	27.5	27.1	26.7	26.6	26.5	26.2	25.1	24.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	28.2	28.9	29.4	29.4	28.8	28.7	28.5	28.3	28.5	28.1	27.6		
		MCDB	32.2	33.1	34.2	35.2	34.4	32.9	33.1	32.4	32.1	32.5	32.6	32.8		
	2%	WB	26.7	27.5	28.1	28.7	28.7	28.2	28.1	27.7	27.7	27.6	27.2	26.7		
		MCDB	30.5	31.7	32.8	33.7	33.1	31.9	31.6	31.2	31.1	31.4	31.5	31.3		
	5%	WB	26.3	26.9	27.6	28.3	28.3	27.9	27.6	27.3	27.2	27.1	26.7	26.1		
		MCDB	29.8	30.7	31.9	32.8	32.3	31.4	30.9	30.6	30.4	30.7	30.8	30.3		
	10%	WB	25.8	26.5	27.2	27.9	27.9	27.5	27.2	27.0	26.8	26.7	26.2	25.5		
		MCDB	29.2	30.0	31.0	32.1	31.6	31.0	30.5	30.2	29.9	30.0	29.9	29.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	6.9	5.9	5.9	5.4	4.9	4.6	4.6	5.2	6.5	8.0	9.0		
		MCDBR	9.8	8.3	7.6	7.4	6.7	6.3	6.4	6.2	6.7	8.0	9.1	9.9		
	5% WB	MCWBR	4.7	3.8	3.2	3.0	3.0	2.9	2.9	2.8	3.0	3.6	4.0	4.6		
		MCWBR	8.1	6.8	6.7	6.6	6.1	5.7	5.7	5.4	5.9	7.2	8.2	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.544	0.594	0.600	0.541	0.478	0.463	0.461	0.461	0.469	0.527	0.489	0.478		
		taud	1.992	1.872	1.907	2.084	2.311	2.373	2.370	2.361	2.321	2.124	2.216	2.211		
	Ebn at Noon		762	741	748	788	825	828	832	843	842	787	807	811		
		Edh at Noon	176	205	202	169	132	122	123	126	132	158	140	139		
(44) All-Sky Solar Radiation	RadAvg		5.04	5.41	5.54	5.78	5.10	4.92	4.76	4.89	4.73	4.79	5.00	4.97		
	RadStd		0.34	0.29	0.40	0.38	0.33	0.28	0.30	0.33	0.29	0.34	0.37	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.32	-0.32	N/A	N/A	N/A	-184	-184
(47) Regional (24 neighbors)		N/A	N/A	N/A	+0.30	N/A	+0.18	N/A	N/A	+72	+72

Nomenclature: See separate page